# Midterm Project Report

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## Introduction:

For my midterm project, I chose to answer the question: *Do students with higher GPAs report experiencing more mental health issues like depression and anxiety?* I selected this topic because college plays a major role in mental health, and many students, including myself and my peers can relate to the stress and pressure that comes with academic performance. I was particularly interested in whether students who excel academically can also maintain good mental health, as this topic is not often talked about. Additionally, I hope this project raises awareness about mental health and encourages a healthier balance between academics and mental well-being.

To answer my research question, I used the *Student Mental Health* dataset by Shariful Islam, which I obtained through Kaggle. The dataset was collected through a survey conducted at the University of Malaysia and consists of 101 observations across 11 variables, such as gender, age, and major. I chose this dataset because it directly includes self-reported mental health indicators (depression and anxiety) as well as academic performance (GPA), making it highly relevant for analyzing the relationship between mental health and student achievement. For my analysis, I focused on three variables: depression, anxiety, and GPA. The depression and anxiety variables are represented as “Yes” or “No”, while GPA is categorized into ranges (2.50 - 2.99, 3.00 - 3.49, etc.).

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.4     ✔ readr     2.1.5
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ ggplot2   3.5.1     ✔ tibble    3.2.1
## ✔ lubridate 1.9.4     ✔ tidyr     1.3.1
## ✔ purrr     1.0.4
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

```
df <- read_csv("Student Mental health.csv")
```

```
## Rows: 101 Columns: 11
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr (10): Timestamp, Choose your gender, What is your course?, Your current ...
## dbl  (1): Age
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

# Data analysis:

For this project, I performed descriptive analysis to examine the relationship between GPA and mental health variables (depression and anxiety). I renamed the three columns used to answer my question for clarity and to make the coding process easier. Then, I converted GPA from a character to a categorical variable. After this, I generated frequency tables for the depression and anxiety variables to analyze how many students in each GPA range reported experiencing these mental health issues. To visualize my findings, I generated bar plots, which helped me compare the distribution of depression and anxiety across the GPA categories.

```
#to explore
dim(df)
```

```
## [1] 101  11
```

```
str(df)
```

```
## spc_tbl_ [101 × 11] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Timestamp                                  : chr [1:101] "8/7/2020 12:02" "8/7/20
20 12:04" "8/7/2020 12:05" "8/7/2020 12:06" ...
##  $ Choose your gender                         : chr [1:101] "Female" "Male" "Male"
"Female" ...
##  $ Age                                        : num [1:101] 18 21 19 22 23 19 23 18
19 18 ...
##  $ What is your course?                       : chr [1:101] "Engineering" "Islamic e
ducation" "BIT" "Laws" ...
##  $ Your current year of Study                 : chr [1:101] "year 1" "year 2" "Year
1" "year 3" ...
##  $ What is your CGPA?                         : chr [1:101] "3.00 - 3.49" "3.00 - 3.
49" "3.00 - 3.49" "3.00 - 3.49" ...
##  $ Marital status                             : chr [1:101] "No" "No" "No" "Yes" ...
##  $ Do you have Depression?                    : chr [1:101] "Yes" "No" "Yes" "Yes"
...
##  $ Do you have Anxiety?                       : chr [1:101] "No" "Yes" "Yes" "No"
...
##  $ Do you have Panic attack?                  : chr [1:101] "Yes" "No" "Yes" "No"
...
##  $ Did you seek any specialist for a treatment?: chr [1:101] "No" "No" "No" "No" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   Timestamp = col_character(),
##   ..   `Choose your gender` = col_character(),
##   ..   Age = col_double(),
##   ..   `What is your course?` = col_character(),
##   ..   `Your current year of Study` = col_character(),
##   ..   `What is your CGPA?` = col_character(),
##   ..   `Marital status` = col_character(),
##   ..   `Do you have Depression?` = col_character(),
##   ..   `Do you have Anxiety?` = col_character(),
##   ..   `Do you have Panic attack?` = col_character(),
##   ..   `Did you seek any specialist for a treatment?` = col_character()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```
head(df)
```

```
## # A tibble: 6 × 11
##   Timestamp      `Choose your gender`   Age `What is your course?`
##   <chr>          <chr>                <dbl> <chr>
## 1 8/7/2020 12:02 Female                  18 Engineering
## 2 8/7/2020 12:04 Male                    21 Islamic education
## 3 8/7/2020 12:05 Male                    19 BIT
## 4 8/7/2020 12:06 Female                  22 Laws
## 5 8/7/2020 12:13 Male                    23 Mathemathics
## 6 8/7/2020 12:31 Male                    19 Engineering
## # ℹ 7 more variables: `Your current year of Study` <chr>,
## #   `What is your CGPA?` <chr>, `Marital status` <chr>,
## #   `Do you have Depression?` <chr>, `Do you have Anxiety?` <chr>,
## #   `Do you have Panic attack?` <chr>,
## #   `Did you seek any specialist for a treatment?` <chr>
```

```
tail(df)
```

```
## # A tibble: 6 × 11
##   Timestamp           `Choose your gender`   Age `What is your course?`
##   <chr>               <chr>                <dbl> <chr>
## 1 13/07/2020 19:08:32 Female                  18 Islamic Education
## 2 13/07/2020 19:56:49 Female                  21 BCS
## 3 13/07/2020 21:21:42 Male                    18 Engineering
## 4 13/07/2020 21:22:56 Female                  19 Nursing
## 5 13/07/2020 21:23:57 Female                  23 Pendidikan Islam
## 6 18/07/2020 20:16:21 Male                    20 Biomedical science
## # ℹ 7 more variables: `Your current year of Study` <chr>,
## #   `What is your CGPA?` <chr>, `Marital status` <chr>,
## #   `Do you have Depression?` <chr>, `Do you have Anxiety?` <chr>,
## #   `Do you have Panic attack?` <chr>,
## #   `Did you seek any specialist for a treatment?` <chr>
```

```
sum(is.na(df))
```

```
## [1] 1
```

```
summary(df)
```

```
##   Timestamp         Choose your gender      Age        What is your course?
##  Length:101         Length:101         Min.   :18.00   Length:101
##  Class :character   Class :character   1st Qu.:18.00   Class :character
##  Mode  :character   Mode  :character   Median :19.00   Mode  :character
##                                        Mean   :20.53
##                                        3rd Qu.:23.00
##                                        Max.   :24.00
##                                        NA's   :1
##  Your current year of Study What is your CGPA? Marital status
##  Length:101                 Length:101         Length:101
##  Class :character           Class :character   Class :character
##  Mode  :character           Mode  :character   Mode  :character
##
##
##
##
##  Do you have Depression? Do you have Anxiety? Do you have Panic attack?
##  Length:101              Length:101           Length:101
##  Class :character        Class :character     Class :character
##  Mode  :character        Mode  :character     Mode  :character
##
##
##
##
##  Did you seek any specialist for a treatment?
##  Length:101
##  Class :character
##  Mode  :character
##
##
##
##
```

```
df <- df |>
  rename("depression" = "Do you have Depression?", "anxiety" = "Do you have Anxiety?",
"gpa" = "What is your CGPA?")
df
```

```
## # A tibble: 101 × 11
##    Timestamp      `Choose your gender`   Age `What is your course?`
##    <chr>          <chr>                <dbl> <chr>
##  1 8/7/2020 12:02 Female                  18 Engineering
##  2 8/7/2020 12:04 Male                    21 Islamic education
##  3 8/7/2020 12:05 Male                    19 BIT
##  4 8/7/2020 12:06 Female                  22 Laws
##  5 8/7/2020 12:13 Male                    23 Mathemathics
##  6 8/7/2020 12:31 Male                    19 Engineering
##  7 8/7/2020 12:32 Female                  23 Pendidikan islam
##  8 8/7/2020 12:33 Female                  18 BCS
##  9 8/7/2020 12:35 Female                  19 Human Resources
## 10 8/7/2020 12:39 Male                    18 Irkhs
## # ℹ 91 more rows
## # ℹ 7 more variables: `Your current year of Study` <chr>, gpa <chr>,
## #   `Marital status` <chr>, depression <chr>, anxiety <chr>,
## #   `Do you have Panic attack?` <chr>,
## #   `Did you seek any specialist for a treatment?` <chr>
```

```
df$gpa <- as.factor(df$gpa)

str(df)
```

```
## spc_tbl_ [101 × 11] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Timestamp                                    : chr [1:101] "8/7/2020 12:02" "8/7/20
20 12:04" "8/7/2020 12:05" "8/7/2020 12:06" ...
##  $ Choose your gender                           : chr [1:101] "Female" "Male" "Male"
"Female" ...
##  $ Age                                          : num [1:101] 18 21 19 22 23 19 23 18
19 18 ...
##  $ What is your course?                         : chr [1:101] "Engineering" "Islamic e
ducation" "BIT" "Laws" ...
##  $ Your current year of Study                   : chr [1:101] "year 1" "year 2" "Year
1" "year 3" ...
##  $ gpa                                          : Factor w/ 5 levels "0 - 1.99","2.00
- 2.49",..: 4 4 4 4 4 5 5 5 3 5 ...
##  $ Marital status                               : chr [1:101] "No" "No" "No" "Yes" ...
##  $ depression                                   : chr [1:101] "Yes" "No" "Yes" "Yes"
...
##  $ anxiety                                      : chr [1:101] "No" "Yes" "Yes" "No"
...
##  $ Do you have Panic attack?                    : chr [1:101] "Yes" "No" "Yes" "No"
...
##  $ Did you seek any specialist for a treatment?: chr [1:101] "No" "No" "No" "No" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   Timestamp = col_character(),
##   ..   `Choose your gender` = col_character(),
##   ..   Age = col_double(),
##   ..   `What is your course?` = col_character(),
##   ..   `Your current year of Study` = col_character(),
##   ..   `What is your CGPA?` = col_character(),
##   ..   `Marital status` = col_character(),
##   ..   `Do you have Depression?` = col_character(),
##   ..   `Do you have Anxiety?` = col_character(),
##   ..   `Do you have Panic attack?` = col_character(),
##   ..   `Did you seek any specialist for a treatment?` = col_character()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```
tab <- table(df$gpa, df$depression)
tab
```

```
##
##               No Yes
##   0 - 1.99     4   0
##   2.00 - 2.49  2   0
##   2.50 - 2.99  1   3
##   3.00 - 3.49 24  19
##   3.50 - 4.00 35  13
```

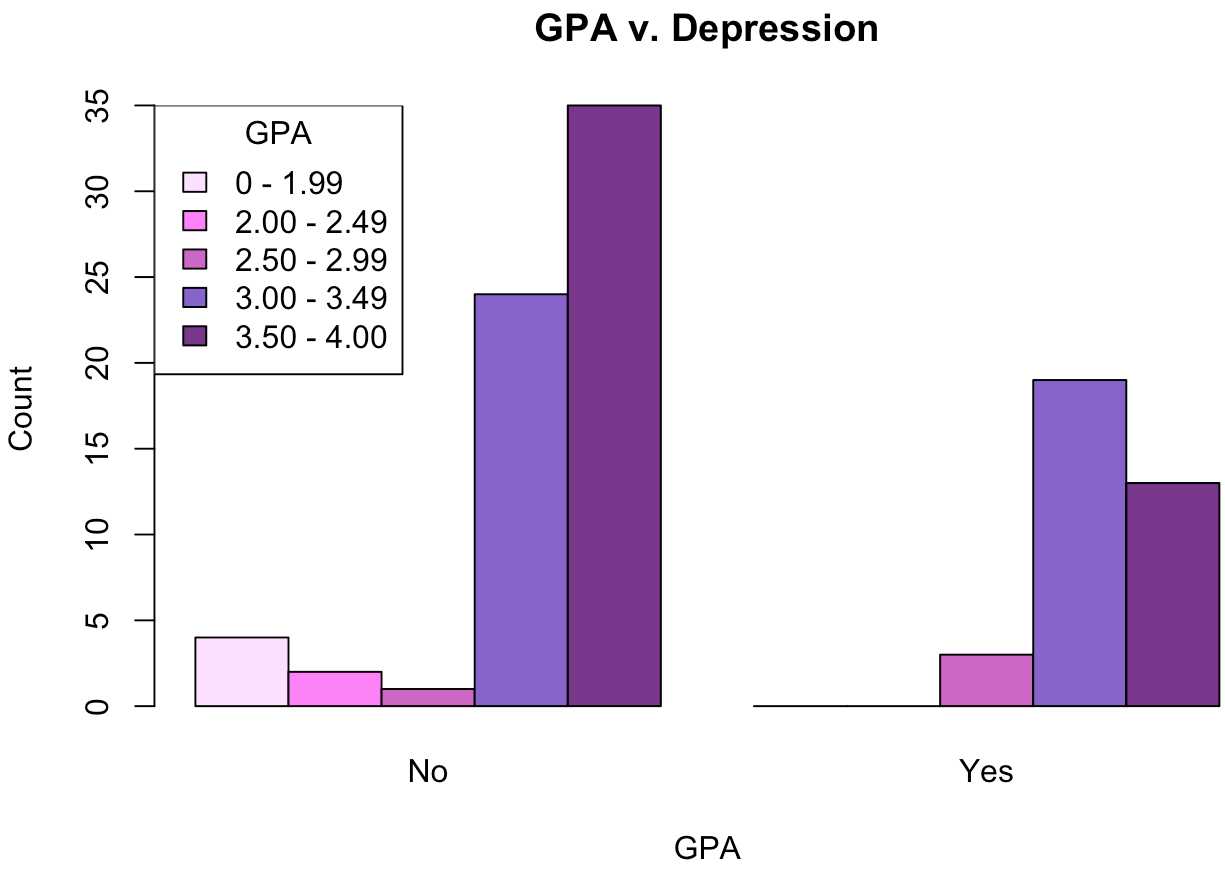
```
tab2 <- table(df$gpa, df$anxiety)
tab2
```

```
##
##               No Yes
##   0 - 1.99     4   0
##   2.00 - 2.49  2   0
##   2.50 - 2.99  3   1
##   3.00 - 3.49 28  15
##   3.50 - 4.00 30  18
```

```
df_col<- c("thistle1", "orchid1", "orchid3","mediumpurple3", "mediumorchid4")

barplot(tab, beside = TRUE, col = df_col,
        main = "GPA v. Depression",
        xlab = "GPA",
        ylab = "Count")

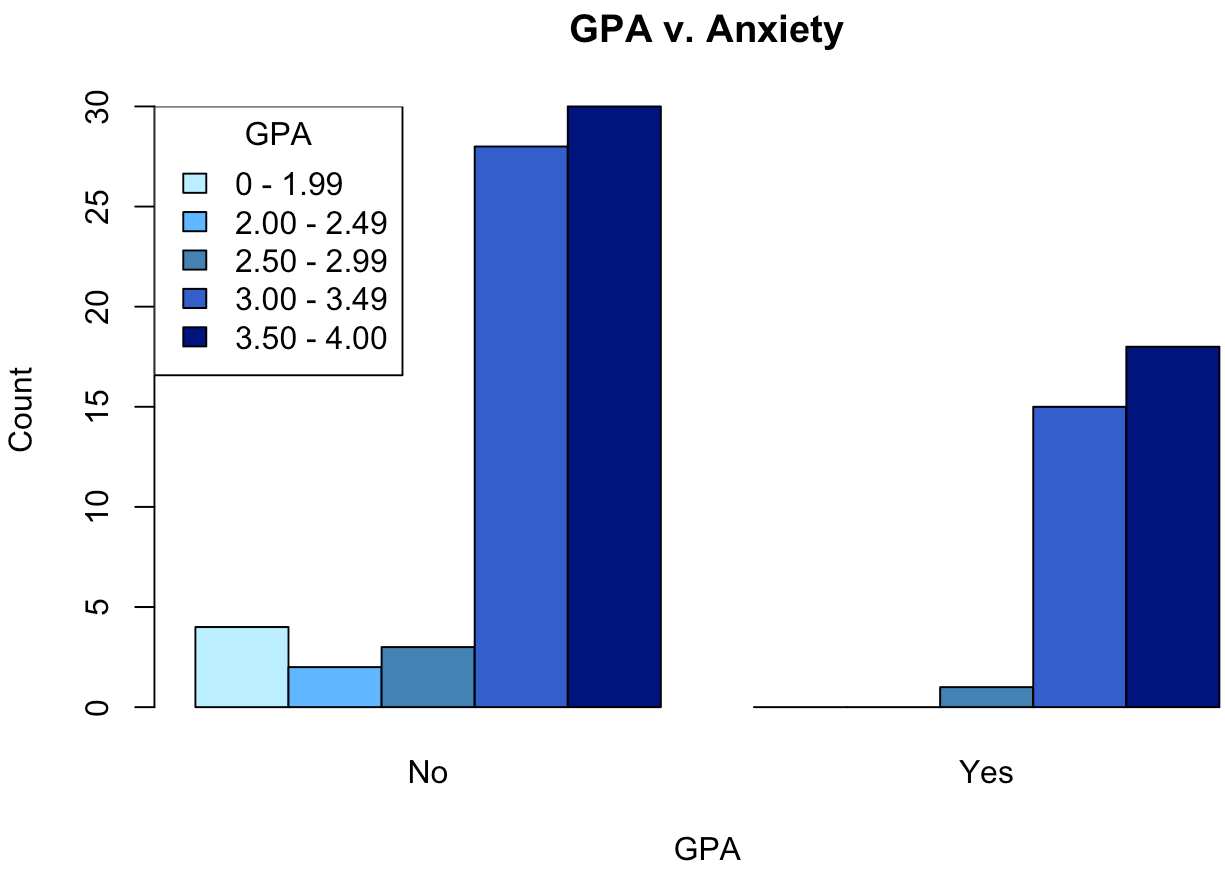
legend("topleft", legend = rownames(tab), fill = df_col, title = "GPA", horiz = FALSE)
```



```
df_col<- c("lightblue1", "steelblue1", "steelblue","royalblue3", "navy")

bp <- barplot(tab2, beside = TRUE, col = df_col,
        main = "GPA v. Anxiety",
        xlab = "GPA",
        ylab = "Count")

legend("topleft", legend = rownames(tab2), fill = df_col, title = "GPA", horiz = FALSE)
```




# Findings

Through my analysis, I found that students with higher GPAs do report experiencing mental health issues, however, the patterns vary between depression and anxiety. The data shows that students with a GPA between 3.00 - 3.49 reported the highest levels of depression, while those with a GPA of 3.50 - 4.00 reported the highest levels of anxiety. Surprisingly, students with lower GPAs (below 2.99) had significantly fewer reports of depression and anxiety. These findings suggest that academic success may come with higher mental health struggles, possibly due to increased academic pressures and stress.

These results highlight the need for better mental health support, especially for high-achieving students who may be struggling in ways that are not always easy to tell. Future research could explore additional variables that influence this relationship, such as students' majors age, and gender. Additionally, expanding the dataset to include students from different universities and community colleges would increase the sample size and diversity, leading to more accurate results and clearer trends.

# References:

Islam, Shariful. (2021). *Student Mental Health Dataset*. https://www.kaggle.com/datasets/shariful07/student-mental-health (https://www.kaggle.com/datasets/shariful07/student-mental-health). Accessed March 2, 2025.